Windows Virtualization Report/Guide

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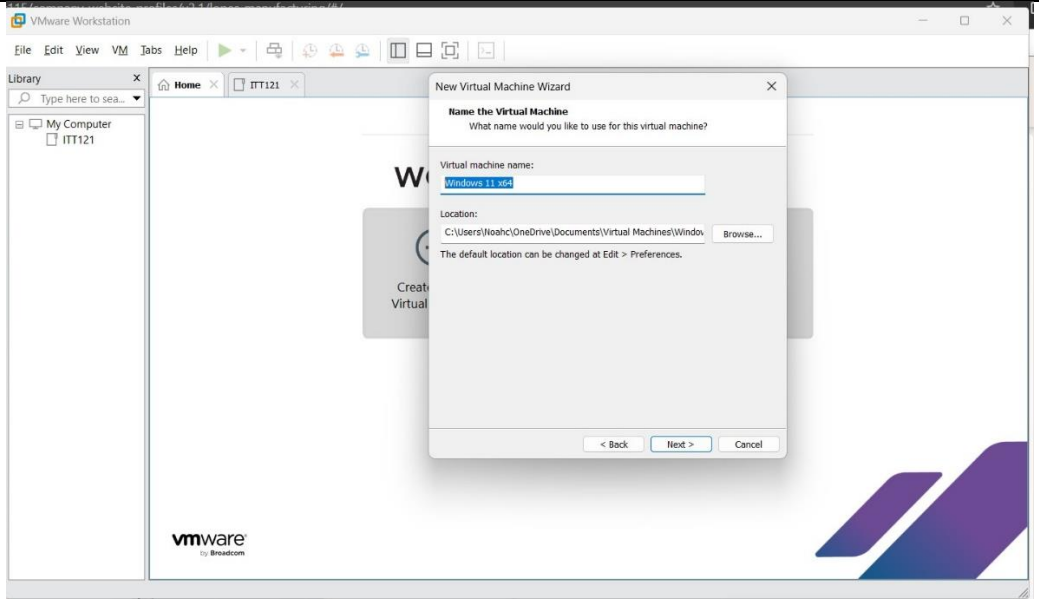
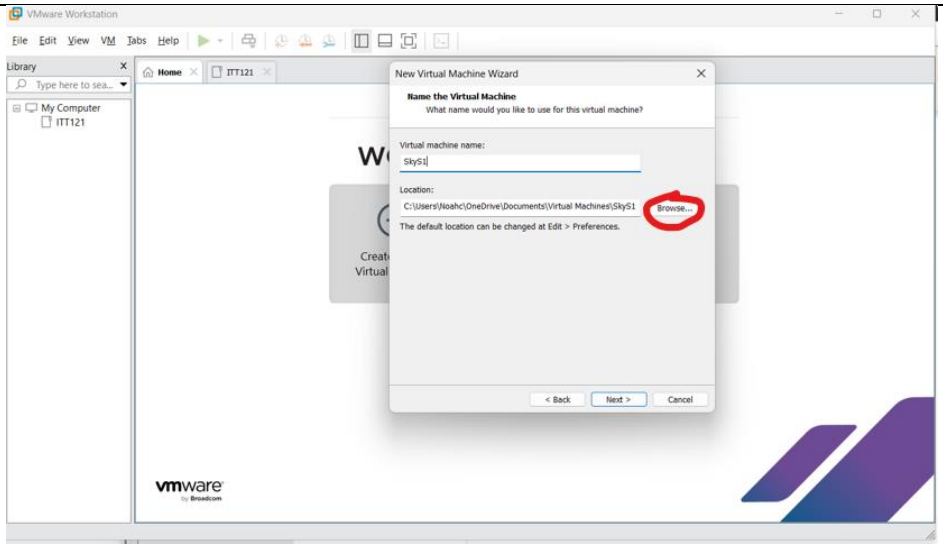
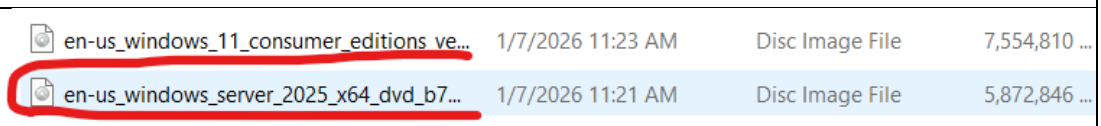
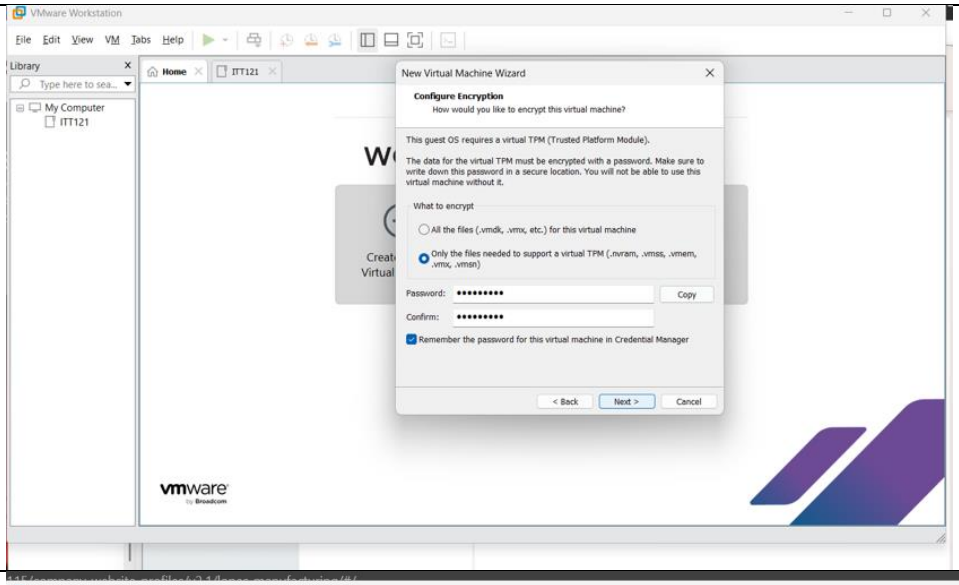
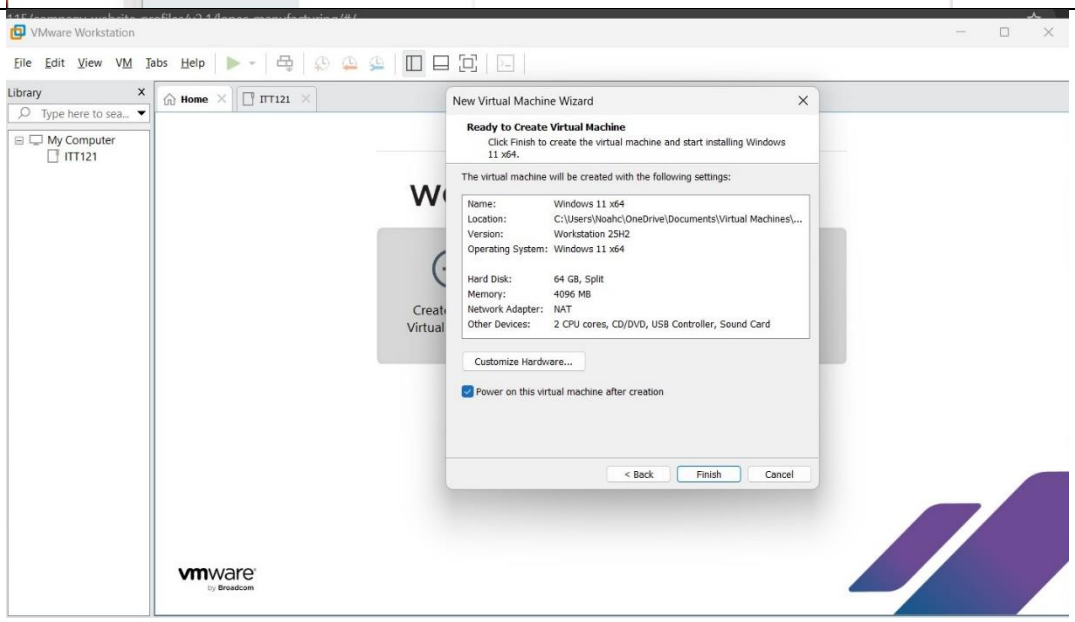
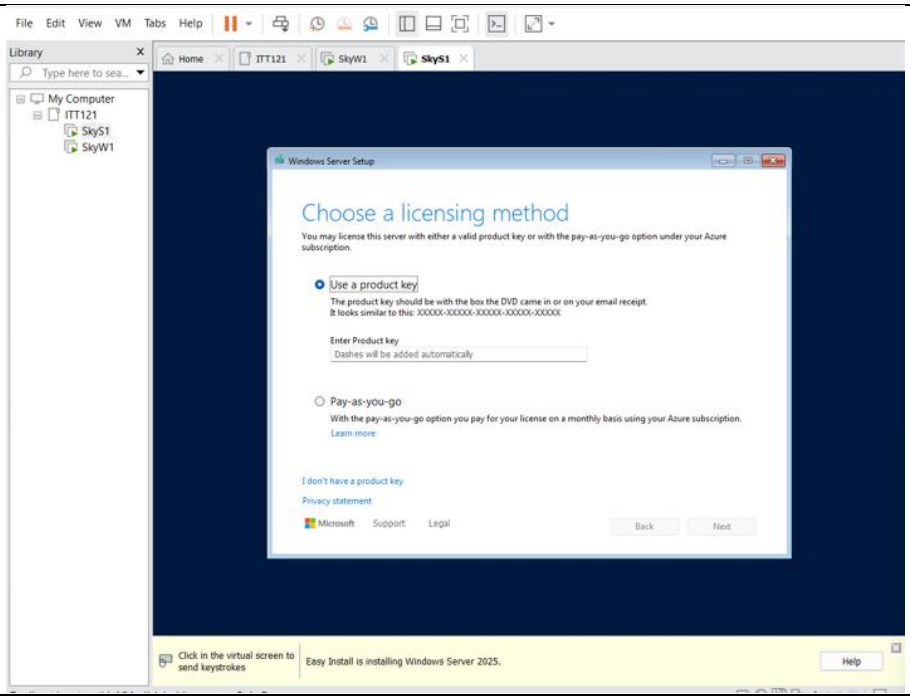
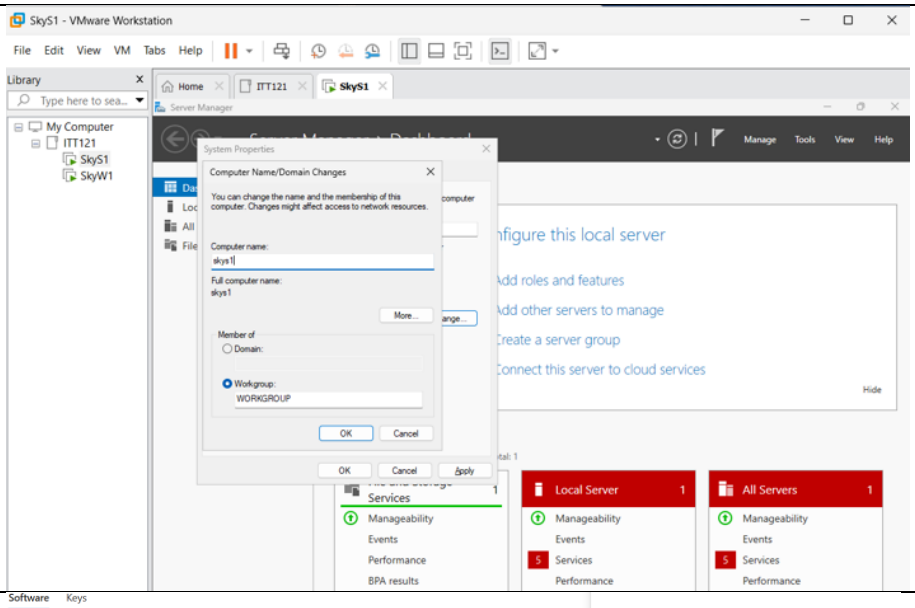
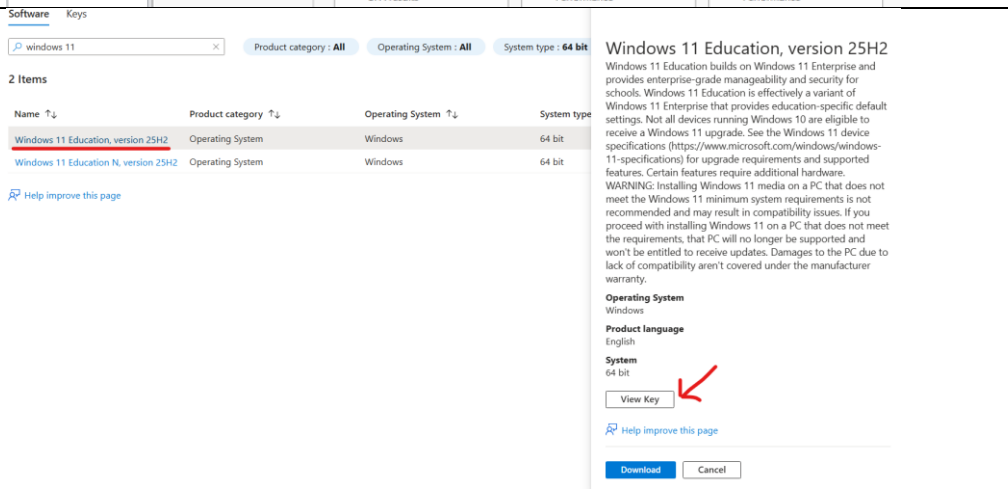
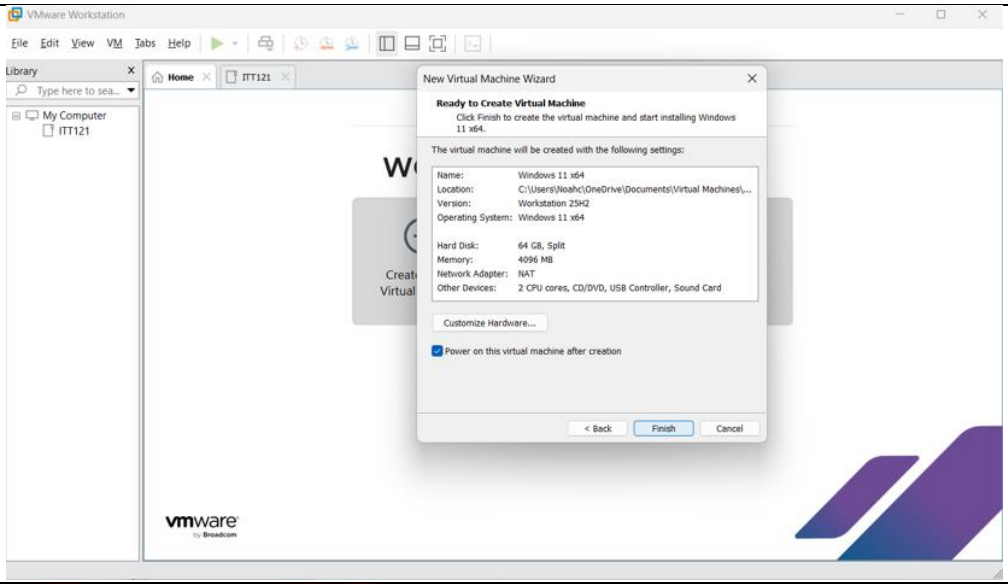
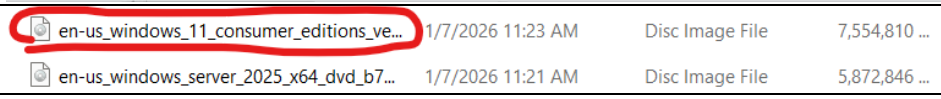
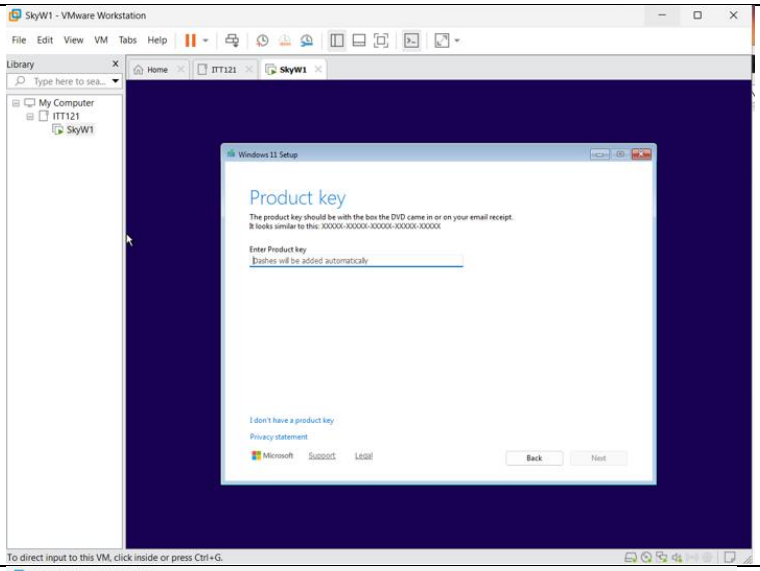
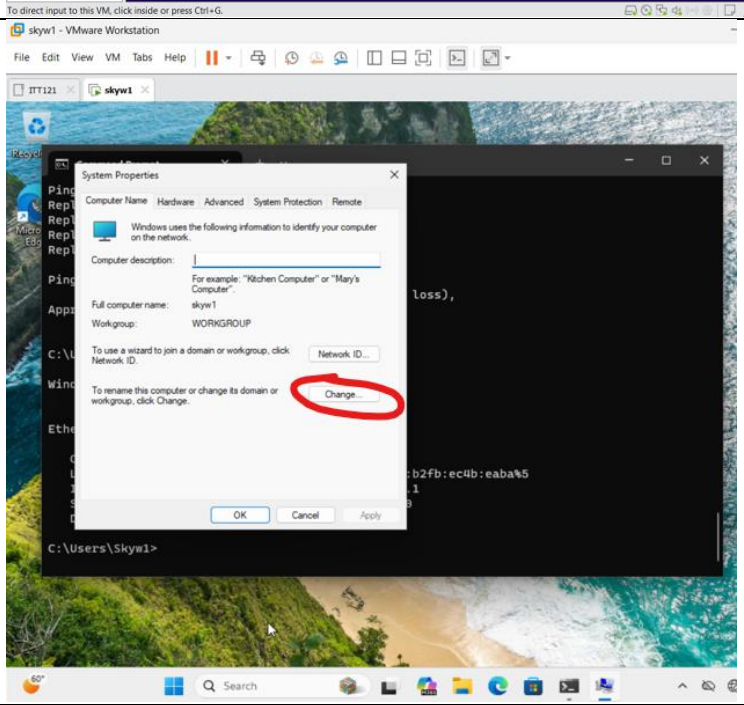
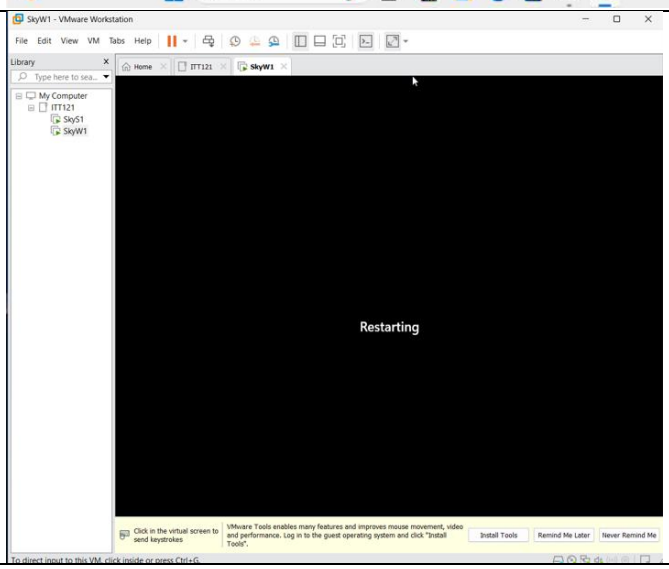
Within VMware, create a new machine and name it appropriately.	
Before you continue, you will need to select the correct ISO so that VMware will actually create Windows Server.	
Open file explorer and in your downloads, find the circled file. The file should be named something along the lines of “windows_server_2025 ". Once you select the file, you will be brought back to VMware installation where you can continue.	

For the server installation, you will have the option to set a password during the VMware setup. Set the password as “Red.vine1” for both input boxes. Continue through the rest of the installation.	
Before you select finish, make sure that you have the correct amount of hard disk space allocated (64GB), correct amount of memory (4096MB), and the correct amount of cores (2 CPU cores).	
The machine will now launch and you can work your way through the installation, the most demanding step in this process is entering the license key when prompted. This is the key we saved from Azure. Enter the key and continue through the installation.	

Once the VM finishes and you are logged into the server, press the windows icon key on your keyboard and type in “sysdm.cpl” and press enter. This will pull up the system properties screen. Choose the “change” button and rename the server appropriately. After this the machine will need to restart.	
No we will repeat this process for Windows 11 with very few differences. Start by finding the ISO on Azure. In the free software, search for “windows 11” and choose **windows 11 education**. Do not choose the N version. Make sure to view and save the key before downloading the image.	
Once the image has finished downloading, create a new VM in VMware and move through the setup, before you finish the setup make sure your disk, memory, and CPU core match what is on screen. (64GB, 4096MB, 2 CPU cores).	
Make sure to choose the correct image	

Once the machine starts up it will once again ask you for the product key, enter this key here before moving on with the installation process.	
Once the machine finishes downloading and you are inside the Windows 11 desktop, use “sysdm.cpl” in the search bar to find system properties. Choose the “change” option and give the workstation an appropriate name.	
After renaming the machine it will need to restart. After this process you will have both the server and windows 11 installed.	

Lopes Manufacturing's mission to design, manufacture, and install high-quality helicopter parts for the U.S. military, will require a secure, and reliable computing environment. Windows Server 2025 and Windows 11 fit naturally into this corporate environment by providing enterprise-grade infrastructure, centralized management, and strong security controls that support mission-critical operations. Windows Server 2025 can be used to host core services such as Active Directory, file and print services, authentication, and internal applications, while Windows 11 serves as the standardized client operating system for engineers, administrators, and production staff.

Modern host virtualization enables Lopes Manufacturing to maximize hardware utilization while maintaining isolation between workloads. Using VMware as the hypervisor, physical server resources such as CPU, memory, storage, and networking are allocated dynamically to virtual machines. Storage virtualization supports snapshots and backups, allowing rapid recovery in the event of system failure. Live migration features further enhance availability by allowing virtual machines to be moved between hosts with minimal downtime.

The operating system plays a critical role by acting as an intermediary between hardware and applications. Windows Server manages hardware resources, enforces security policies, and provides services that applications depend on, while Windows 11 offers a secure and user-friendly interface for end users. Both operating systems interact with virtualized hardware in the same way they would with physical hardware, which simplifies deployment and management.

Operating systems also present security challenges, including malware, unauthorized access, and misconfigurations. Windows addresses these concerns through features such as built-in firewalls, role-based access control, patch management, and encryption. Virtualization further enhances security by isolating workloads and limiting the impact of compromised systems.

Fundamentally, virtualization allows multiple operating systems to run independently on shared hardware. Type 1 hypervisors, such as VMware, run directly on hardware and offer better performance and security than Type 2 hypervisors, which run on top of an existing operating system. For a defense-focused manufacturing company like Lopes Manufacturing, Type 1 virtualization provides the reliability and control required to support sensitive and mission-critical operations.

References

What is windows server?. (Accessed January 17, 2026)Microsoft Learn. https://learn.microsoft.com/en-us/windows-server/get-started/overview